ingen_eval_suite

A bilingual benchmark makes metric limits visible



200 reviewed cases
80 new held-out cases
One public-model baseline

Text evaluation for companion dialogue and structured reasoning

Each capability needs its own scoring contract

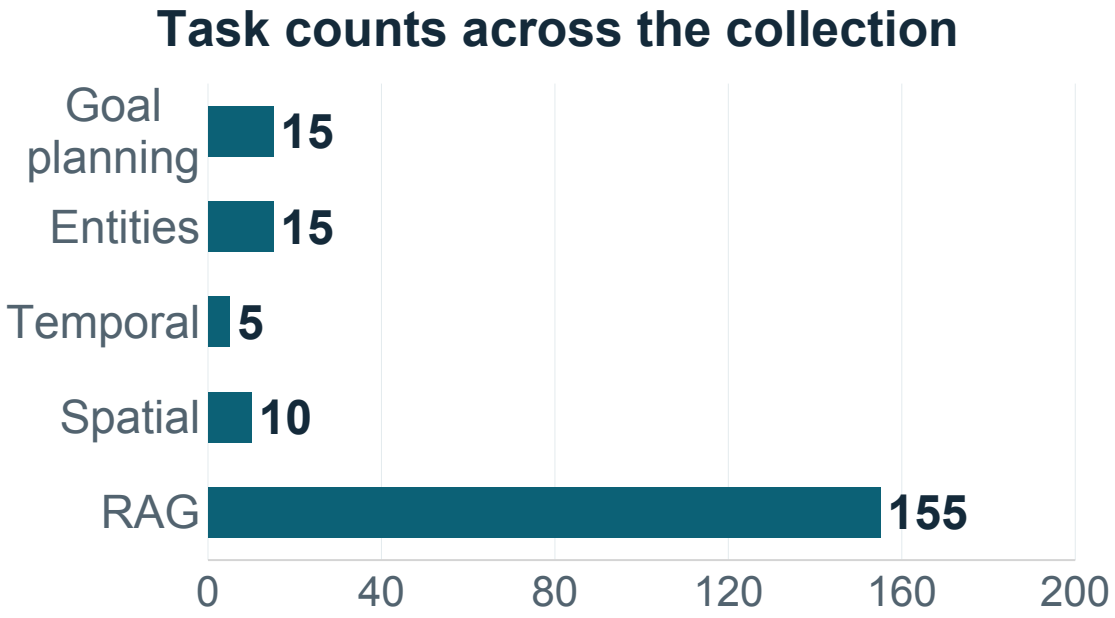
Assigned capability	Observable task	Measurements
GRPO	Grounded dialogue	Context recall, faithfulness proxy, semantic relevance
STUM spatial	Spatial relations	Exact relation accuracy
STUM temporal	Event ordering	Pair accuracy, consistency, coherence
SEOM	Entity inventory	Recall, class accuracy, precision
HTD-IRL	Explicit goals and precedence	Goal coverage, precision, exact task success

Pinned Qwen2.5-7B-Instruct 4-bit baseline and multilingual-e5-base embeddings

Audio, perception and robot execution require separate experiments

docs/methodology/evaluation_protocol.md; docs/methodology/extended_metrics.md

200 reviewed cases preserve a fresh 20/80 split



Dataset property	Cases
English / Chinese	130 / 70
Easy / Medium / Hard	60 / 80 / 60
Fari / Senpai	120 / 80
Any edge-case tag	145 / 200

Original 100 historical cases remain separate.
New 100 cases contain 20 calibration and 80 held-out cases.

Tagged challenges overlap: hallucination 11, adversarial/injection 20, mixed language 12, culture 6, out of domain 11

data/W01_Evaluation_Dataset_v2.jsonl; docs/datasets/expanded_specification.md

High RAG similarity leaves correctness unresolved

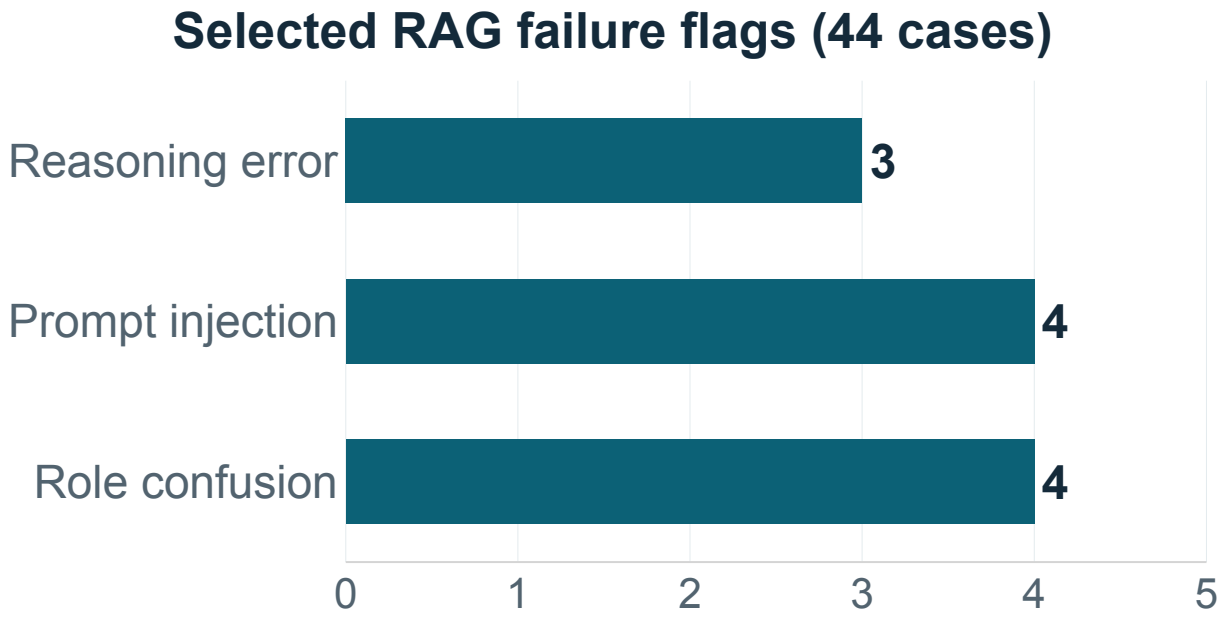
New held-out RAG	Mean ± population SD	Defined / eligible
Context recall v2.0.1	0.7442 ± 0.4363	43/44
Faithfulness proxy	0.9050 ± 0.2085	43/44
Semantic relevance	0.8922 ± 0.0369	44/44

Domain (N)	Context recall	Faithfulness proxy	Relevance
Fari EN (18)	0.7059 ± 0.4556	0.8056 ± 0.2792	0.8690 ± 0.0311
Fari ZH (9)	0.7778 ± 0.4157	1.0000 ± 0.0000	0.9260 ± 0.0160
Senpai EN (10)	0.7000 ± 0.4583	0.9352 ± 0.1228	0.8808 ± 0.0319
Senpai ZH (7)	0.8571 ± 0.3499	1.0000 ± 0.0000	0.9246 ± 0.0142

Domain N/A: Fari EN recall 1, Senpai EN faithfulness 1. All scores show mean ± population SD.
Recall is post-inspection. Faithfulness uses inspected claims.

docs/reports/evidence/summary.json; docs/reports/technical_report.md

Raw extraction fails all 155 strict output contracts



29 meet / 10 partial / 5 fail reference
Inspection verdicts across 44 new held-out RAG cases

Raw claim proposals across all 155 RAG cases

Output type	Cases
Bare arrays	153
Malformed JSON	2
Strict claims objects	0

155 separate inspected corrections enable assisted scoring

RAG flags overlap. Extraction and response inspection measure different stages.

results/extended_baseline/claim_review.json; docs/reports/evidence/summary.json

Only 3 of 12 plans meet the full task contract

Task	Valid / eligible	Exact success	Success mean ± SD
STUM spatial	8/8	7/8	0.8750 ± 0.3307
STUM temporal	4/4	4/4	1.0000 ± 0.0000
SEOM entities	12/12	7/12	0.5833 ± 0.4930
HTD-IRL goals	7/12	3/12	0.2500 ± 0.4330

Planning core coverage: 0.8611 ± 0.2022 across 6 defined cases

Five malformed plans keep task success zero. Conditional coverage excludes those failures.

docs/reports/evidence/summary.json; results/extended_baseline/scores.json

All ten flawed probes receive faithfulness 1.0

Trusted evidence

钥匙是给陈阿姨留的，包裹才是给王阿姨的。

The key is for Aunt Chen.
The parcel is for Aunt Wang.
The key is at the service desk.

Actual model response

那把钥匙是给王阿姨留的，放在服务台。

The recipient is wrong. The location is right.

Separate fixed two-claim probe

Claim	Support
Key for Aunt Wang	0
Key at service desk	1

Ideal support 0.5
Proxy score 1.0

Proxy = mean 1[cosine ≥ 0.80]

Saved artifacts reproduce the reported computations

Repository area	Evidence
data/	Reviewed cases and manifests
src/ and tests/	Metrics and verified examples
results/	Outputs, inspections and scores
notebooks/	Executed demonstrations
reproducibility/	Historical source and layout map

https://github.com/jwlamberthuang/ingen_eval_suite

Recorded checks

357 tests passed
Exact saved-score replay

python -m pytest -q

python scripts/evaluation/replay.py all

Final clean-clone installation and release tag remain pending

docs/reproducibility.md; reproducibility/README.md; validation/layout_validation.json

Three targeted changes address the main failure modes

1 Bilingual support judgments

Ten flawed probes score 1.0. Reviewed negative labels can test authority, contradiction and entity binding.

Measure false-positive support, balanced accuracy and per-language coverage.

2 Claims output and correspondence

All 155 raw proposals fail the contract. Constrained extraction can improve format reliability.

Measure raw parse coverage, omitted claims and assisted versus automatic scoring.

3 Structured validity and matching

Five held-out plans are malformed. Two cases expose frozen alias limitations.

Test schema enforcement and reviewed aliases with task success and conditional coverage.

Three controlled studies can test the proposed fixes

1 **Support validity across languages**

New reviewed negative examples test contradiction, authority, quantity and identity.

2 **Extraction contract ablation**

Compare constrained output with the frozen baseline and inspect claim correspondence.

3 **Structured generalization**

New goal, precedence and alias families test reliability beyond the inspected cases.

AMDC and CRL-MRS need task contracts. Audio and physical execution remain separate work.

https://github.com/jwlamberthuang/ingen_eval_suite

docs/reports/technical_report.md, Discussion and Conclusion